# Killing Rate vs. Protection: Is the Relation Linear?

**Model:** Ma et al. (2024) two-strain model with the paper's SI parameters · one dose · 22 × 22 × 3 × 2 = 2,904 simulations · same engine and endpoint definitions as the main parameter sweep report.

## Open issues to solve next

- **γ is extended beyond the paper's range.** The paper uses 1.1–1.4; this test uses 1.1–3.1 (group choice) so the shape of the relation can be seen. Inside 1.1–1.4 the neutral line moves by only 0.01 in p and looks straight.
- **“Killing rate” and “protection” are our reading of the question** (definitions below). Other readings are possible, for example protection through antibiotic degradation (φ_max, κ_b).
- **Plain model only.** No extinction floor; at high γ and dose both strains fall far below one cell before regrowing.
- **c, the inhibitor dose and all other parameters are at base.** Test 3 uses α = 0.95 only.

## Base parameters & variation range

**Table 1.** Variation levels. p is set through β_min: p = (1 − β_min)(1 − c·ι).

| Quantity | Meaning | Base | Range | Values | Source |
|---|---|---|---|---|---|
| γ (y-axis) | killing rate: maximum lysis coefficient | 1.38 | 1.1–3.1 | 22 | SI range 1.1–1.4, extended |
| p = 1 − β (x-axis) | protection: fraction of lysis R avoids | 0.031 | 0–0.30 | 22 | largest p at base c is 0.307 |
| α | growth of R relative to S | 0.95 | 0.85, 0.90, 0.95 | 3 | inside SI range |
| a0 | antibiotic dose | 2.15 | 2.15 and 100 | 2 | base dose and top of SI range |

## Decoupling of the parameters

- **Full factorial:** every γ is combined with every p, for each α and dose. Each axis has 21 evenly spaced values plus the base value.
- **Only private protection changes along the x-axis.** p is varied through β_min; c is fixed, so R's antibiotic degradation (φ) is the same in every run.

## Sweep setup:

- Per-capita lysis rate of S = γ · H(a) · g(s). Per-capita lysis rate of R = (1 − p) · γ · H(a) · g(s).
- Subtracting the per-capita growth rates of the two strains gives an exact identity: d ln(R/S)/dt = g(s) · [γ · H(a) · p − (1 − α)]. R gains at any moment when γ · H · p > 1 − α.
- Neutral line: for each γ, the protection p* at which R neither gains nor loses share, found by root-finding on the simulation (not read off the grid).
- Two read-outs: after full regrowth (steady state), and during treatment (a0 = 100, t = 100).
- Linearity is judged by the $R^2$ of a straight-line fit and by the largest deviation from that line.

# 1. Lysis Rate of R vs. Protection

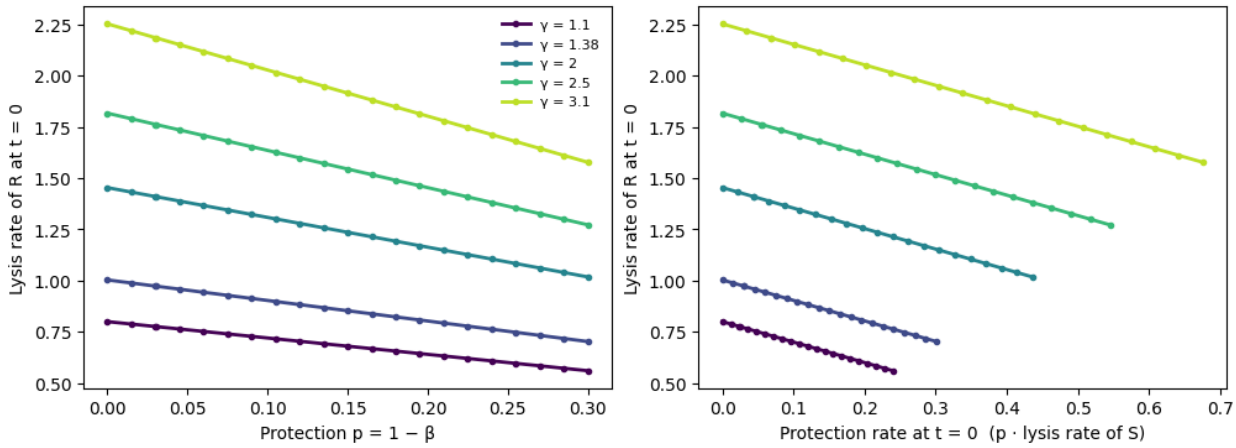



**Figure 1.** Lysis rate of R at the start of treatment against protection (left) and against the protection rate, p × lysis rate of S (right). α = 0.95, base dose.

- Exactly linear for every γ ($R^2 = 1$). Slope = −γ · H(a0) · g(s0), for example −1.00 at γ = 1.38 and −2.25 at γ = 3.1.
- Against the protection rate the slope is −1 for every γ: lysis of R = lysis of S − protection rate.
- This is how the model is built, not a finding.

# 2. Neutral Line Between Killing Rate and Protection

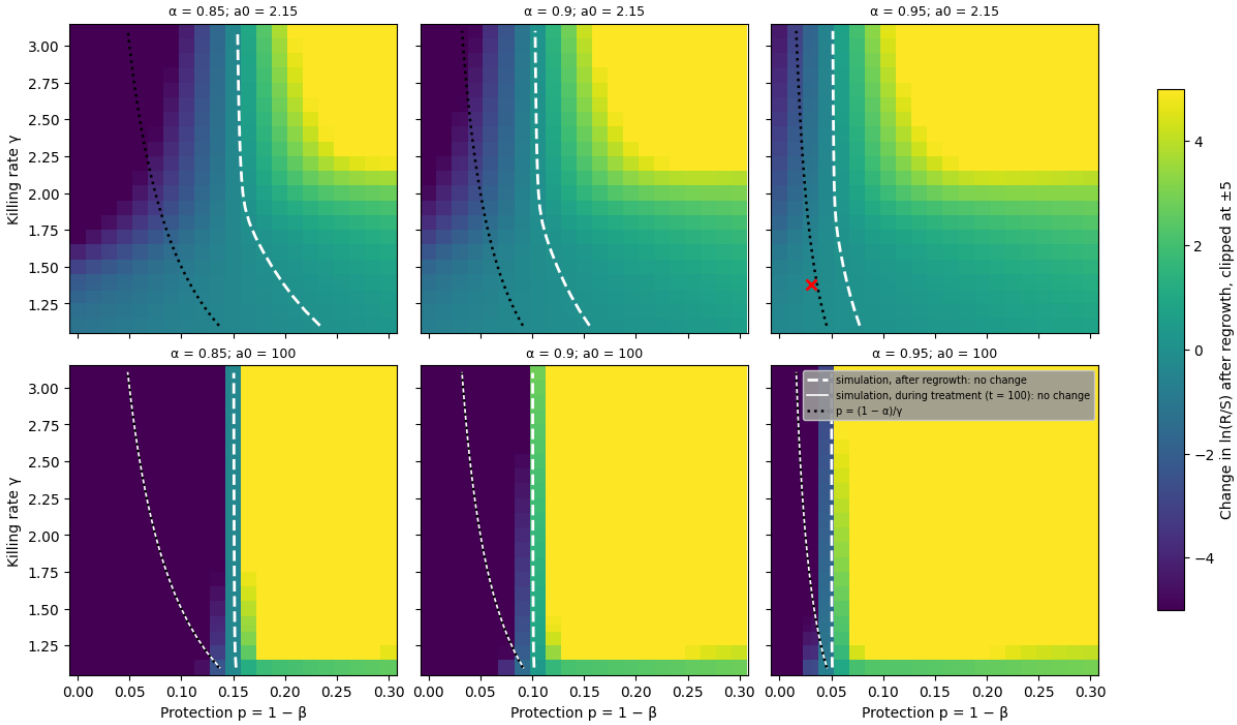



**Figure 2.** Change in ln(R/S) after regrowth on the γ × p plane (above 0 = R gained). White dashed: no change after regrowth. Black dotted: p = (1 − α)/γ. Thin white (bottom row): no change during treatment. Red cross: base point.

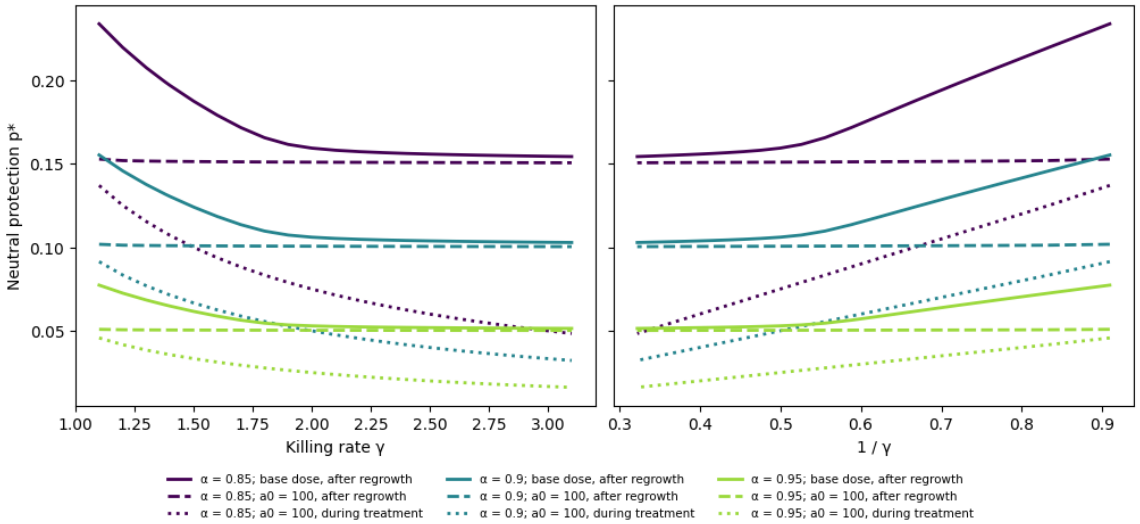



**Figure 3.** Neutral protection p* against γ (left) and against 1/γ (right).

**Table 2.** Straight-line fits of the neutral protection p*. Deviation = largest distance from the fitted line, as a share of the range of p*.

| Dose, read-out | α | p* at γ = 1.1 | p* at γ = 3.1 | $R^2$ vs γ | Deviation vs γ | $R^2$ vs 1/γ | Largest $\lvert p^* - (1-\alpha)/\gamma \rvert$ |
|---|---|---|---|---|---|---|---|
| 100, during treatment | 0.85 | 0.137 | 0.048 | 0.93 | 20% | 1.0000 | 0.0007 |
| 100, during treatment | 0.9 | 0.091 | 0.032 | 0.93 | 20% | 1.0000 | 0.0005 |
| 100, during treatment | 0.95 | 0.046 | 0.016 | 0.93 | 21% | 1.0000 | 0.0003 |
| 100, after regrowth | 0.85 | 0.153 | 0.151 | 0.74 | 46% | 0.9028 | 0.1022 |
| 100, after regrowth | 0.9 | 0.102 | 0.100 | 0.76 | 44% | 0.9163 | 0.0682 |
| 100, after regrowth | 0.95 | 0.051 | 0.050 | 0.77 | 43% | 0.9267 | 0.0341 |
| 2.15, after regrowth | 0.85 | 0.234 | 0.154 | 0.73 | 37% | 0.9236 | 0.1060 |
| 2.15, after regrowth | 0.9 | 0.155 | 0.103 | 0.72 | 38% | 0.9173 | 0.0706 |
| 2.15, after regrowth | 0.95 | 0.077 | 0.051 | 0.71 | 39% | 0.9112 | 0.0353 |

## 2.1 During treatment

- **The relation is inverse, not linear:** $p^* = (1 - \alpha)/\gamma$. The simulation is within 0.0007 of it.
- A straight line against 1/γ ($R^2 = 1.0000$); a curve against γ ($R^2 = 0.93$, up to 21% of the range off a straight line).
- Doubling the killing rate halves the protection R needs to keep its share.

## 2.2 After regrowth

- **The killing rate almost stops mattering.** At a0 = 100, p* stays at 0.050–0.051 from γ = 1.1 to 3.1 (α = 0.95).
- At the base dose p* bends from 0.077 (γ = 1.1) to 0.051 (γ = 3.1), then flattens.
- In every read-out p* is close to proportional to the cost of resistance, 1 − α (within 1.5%).

**Table 3.** Neutral protection after regrowth: simulation vs. the exact formula $p^* = (1 - \alpha)/(1 - \Delta\ln S/G)$, $\alpha = 0.95$. G = total growth $\int g\,dt$; $\Delta\ln S = \ln(\text{final } S / \text{starting } S)$.

| Dose a0 | γ | p* (simulation) | p* (formula) | Total growth G | ΔlnS | (1 − α)/γ |
|---|---|---|---|---|---|---|
| 2.15 | 1.1 | 0.0774 | 0.0774 | 6 | 2.28 | 0.0455 |
| 2.15 | 1.38 | 0.0655 | 0.0655 | 10 | 2.27 | 0.0362 |
| 2.15 | 2 | 0.0530 | 0.0530 | 42 | 2.33 | 0.0250 |
| 2.15 | 3.1 | 0.0514 | 0.0514 | 85 | 2.37 | 0.0161 |
| 100 | 1.1 | 0.0509 | 0.0509 | 122 | 2.13 | 0.0455 |
| 100 | 1.38 | 0.0505 | 0.0505 | 232 | 2.33 | 0.0362 |
| 100 | 2 | 0.0503 | 0.0503 | 357 | 2.36 | 0.0250 |
| 100 | 3.1 | 0.0502 | 0.0502 | 563 | 2.37 | 0.0161 |

- Formula and simulation agree to $3 \times 10^{-7}$. G and ΔlnS are taken from the same run, so this is a consistency relation that explains the result, not a prediction.
- Why: over a full kill-and-regrow cycle, total lysis equals total growth minus the small net gain ΔlnS. When killing is strong, G is far larger than ΔlnS, so $p^* \to 1 - \alpha$ whatever γ is.

# 3. Killing Actually Suffered vs. Protection

- **Peak death rate:** fastest net per-capita decline during the run.
- **Log kill:** ln(starting density / lowest density), the depth of the crash.

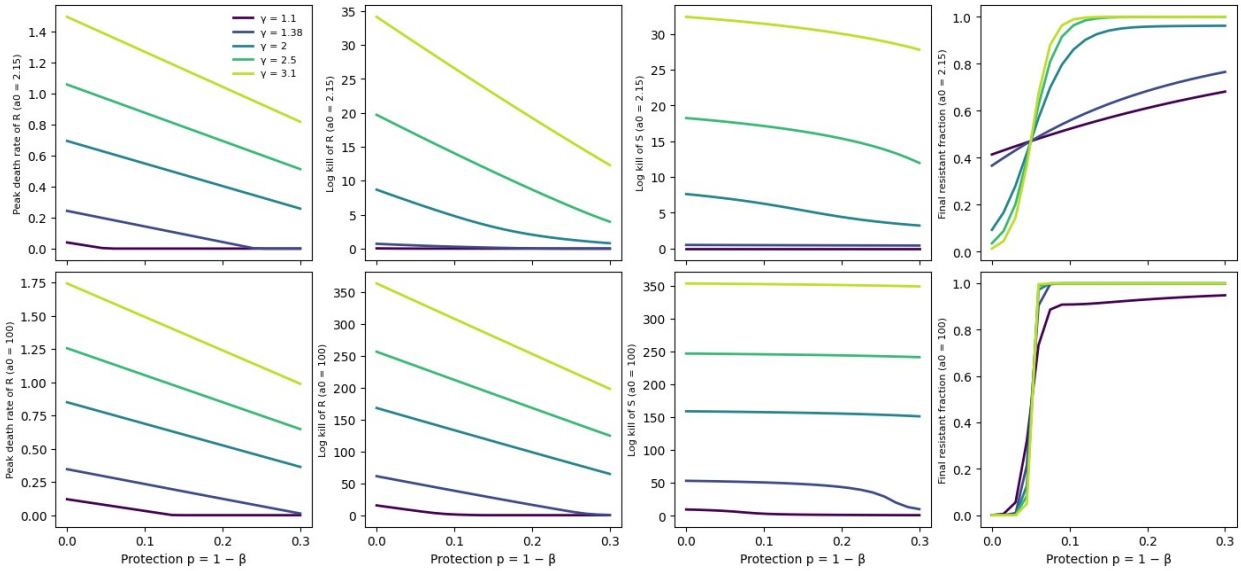



**Figure 4.** Peak death rate of R, log kill of R, log kill of S and final resistant fraction against protection, one line per γ (α = 0.95). Top: base dose. Bottom: a0 = 100.

**Table 4.** $R^2$ of a straight-line fit against protection, by γ (α = 0.95). n/a = S is not killed at all.

| Dose a0 | γ | Lysis rate of R (t = 0) | Peak death rate of R | Log kill of R | Log kill of S | Final f_R |
|---|---|---|---|---|---|---|
| 2.15 | 1.1 | 1.000 | 0.395 | 0.295 | n/a | 0.992 |
| 2.15 | 1.2 | 1.000 | 0.750 | 0.648 | 1.000 | 0.988 |
| 2.15 | 1.38 | 1.000 | 0.980 | 0.904 | 0.995 | 0.973 |
| 2.15 | 1.6 | 1.000 | 1.000 | 0.951 | 0.977 | 0.920 |
| 2.15 | 2 | 1.000 | 1.000 | 0.960 | 0.995 | 0.714 |
| 2.15 | 2.5 | 1.000 | 1.000 | 0.999 | 0.948 | 0.640 |
| 2.15 | 3.1 | 1.000 | 1.000 | 1.000 | 0.972 | 0.600 |
| 100 | 1.1 | 1.000 | 0.773 | 0.651 | 0.811 | 0.580 |
| 100 | 1.2 | 1.000 | 0.940 | 0.869 | 0.940 | 0.524 |
| 100 | 1.38 | 1.000 | 1.000 | 0.996 | 0.754 | 0.525 |
| 100 | 1.6 | 1.000 | 1.000 | 1.000 | 0.879 | 0.518 |
| 100 | 2 | 1.000 | 1.000 | 1.000 | 0.950 | 0.511 |
| 100 | 2.5 | 1.000 | 1.000 | 1.000 | 0.970 | 0.507 |
| 100 | 3.1 | 1.000 | 1.000 | 1.000 | 0.978 | 0.504 |

- **Linear when killing is strong.** Peak death rate of R: $R^2 \geq 0.99$ in 17 of 22 γ levels at the base dose and 20 of 22 at a0 = 100. Log kill of R: 9 of 22 and 19 of 22.
- **Not linear under weaker killing:** γ ≤ 1.4 (peak death rate) and γ ≤ 2.2 (log kill) at the base dose, γ ≤ 1.3 at a0 = 100. There the lines hit zero once R is protected enough to stop declining.

- **Killing of S falls as R's protection rises, and not linearly.** More surviving R cells remove the antibiotic sooner. From p = 0 to 0.30 the log kill of S falls by 5–58% at the base dose; at a0 = 100 it falls up to ten-fold for γ ≤ 1.4 and by less than 5% for γ ≥ 2.0. $R^2 \geq 0.99$ in 5 of 22 levels at the base dose and none at a0 = 100.
- **Final resistant fraction is not linear in protection when killing is strong** (γ ≥ 1.6, or a0 = 100): it is a steep S-shaped switch around p*. At the base dose with γ = 1.1–1.4 it rises smoothly ($R^2$ = 0.97–0.99).

### 3.1 Killing of R vs. killing of S

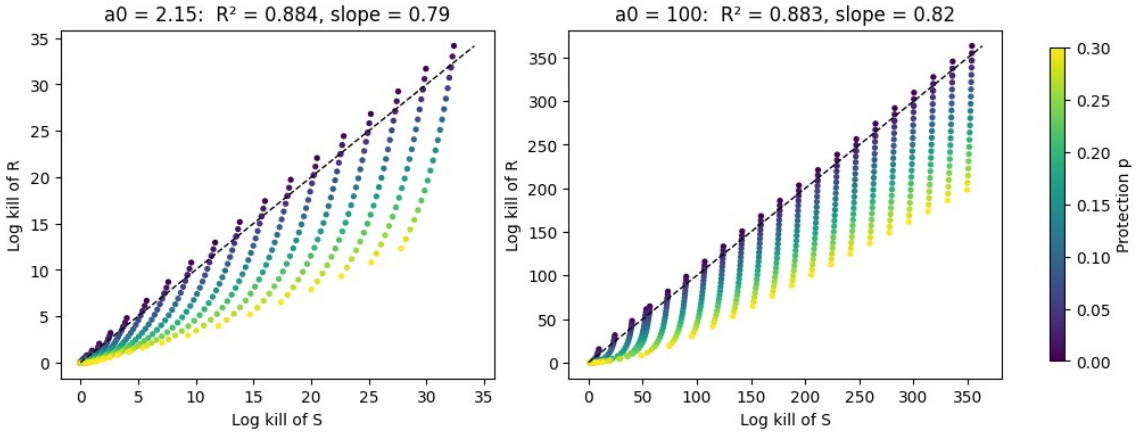



**Figure 5.** Log kill of R against log kill of S, one dot per run (all γ and p, α = 0.95). Dashed: equal killing.

**Table 5.** Straight-line fits of log kill of R against log kill of S.

| Dose a0 | Runs | n | $R^2$ | Slope |
|---|---|---|---|---|
| 2.15 | all runs | 484 | 0.884 | 0.79 |
| 2.15 | p = 0 | 22 | 1.000 | 1.05 |
| 2.15 | p = 0.30 | 22 | 0.989 | 0.43 |
| 100 | all runs | 484 | 0.883 | 0.82 |
| 100 | p = 0 | 22 | 1.000 | 1.01 |
| 100 | p = 0.30 | 22 | 0.985 | 0.58 |

- Without protection the log kill of R runs parallel to that of S (slope 1.01–1.05, $R^2$ = 1.000) but is larger, by 0.03–1.8 ln units at the base dose and 6–10 at a0 = 100, because of R's growth cost.
- Protection lowers the slope: 0.43 (base dose) and 0.58 (a0 = 100) at p = 0.30.
- Over all runs the two are strongly but not perfectly linearly related ($R^2$ = 0.88): each value of p has its own line.

## 4. Conclusions

- **Rates:** lysis of R is exactly linear in protection, by construction.
- **Trade-off during treatment:** killing rate and neutral protection are inversely related, p* = (1 − α)/γ. Linear in 1/γ, not in γ.
- **Trade-off after regrowth:** p* ≈ 1 − α, nearly independent of the killing rate when killing is strong.
- **Killing suffered:** for R, linear in protection only when killing is strong; for S it falls with R's protection but not linearly.

- **Composition:** under strong killing the final resistant fraction switches sharply around p*, so it is not linear in protection.
- Next: repeat with protection through antibiotic degradation (φ_max, κ_b), and with the extinction floor.